DetRefiner: Model-Agnostic Detection Refinement with Feature Fusion Transformer

Soichiro Okazaki, Tatsuya Sasaki, Hiroki Ohashi
Hitachi, Ltd. Research and Development Group

CVPR JUNE 3-7, 2026 DENVER COLORADO

A lightweight plug-and-play refinement module that improves open-vocabulary object detection by fusing global and local features.

Motivation

- Open-Vocabulary Object Detection must recognize both seen and unseen categories.
- Existing detectors often suffer from semantic misalignment and poorly calibrated confidence scores, especially for rare, ambiguous or fine-grained categories.
- Correct detection may receive low scores, while visually plausible false positive can be overconfident.

Why DetRefiner ?

Model-Agnostic and Plug-and-Play
- works on top of existing detectors without internal access.

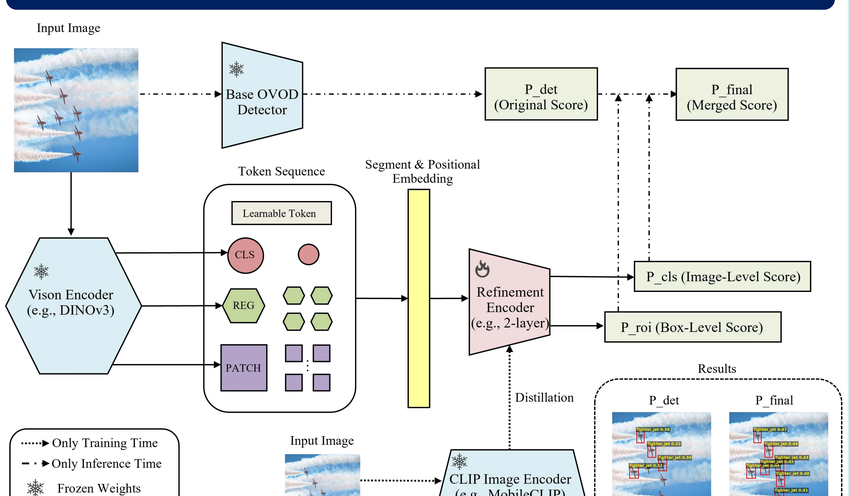
Post-hoc Refinement
- no retraining of the base detectors.

Improvement on score calibration
- for rare and unseen categories.

Key Contribution

- Lightweight fusion of global-local features with a 7.5M-parameter transformer encoder.
- Distillation-enhanced calibration using CLIP-aligned supervision for class/patch vectors.
- Consistent improvements across multiple OVOD detectors and benchmarks.

Method Overview



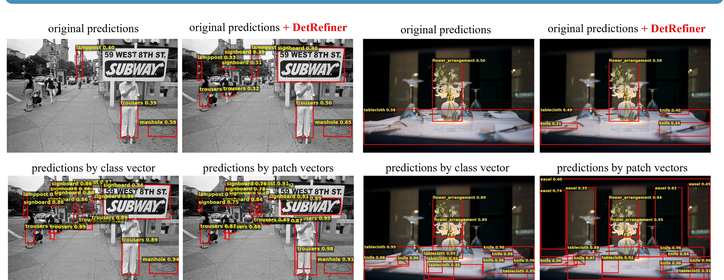
① **Refines** global and patch-level DINOv3 features with a lightweight Refinement Encoder, then pools box-level representations using ROI Align.
② **Matches** class and ROI features with CLIP text embeddings, while using CLIP-based distillation for semantic alignment.
③ **Merges** image-level, box-level, and base detector scores to recalibrate detections without modifying the detector.

Inference Characteristics

- ✓ Use only image and detector outputs (i.e., labels, boxes, confidences).
- ✓ Does not change box-coordinates and category labels.
- ✓ Rescue low-scored true positives and suppress false positives.
- ✓ A single trained DetRefiner generalizes to different detectors without further retraining.

Main Results

(Top) Qualitative Results / (Bottom) Quantitative Results



OV-LVIS: improvements on all 10 reproduced detectors

Model (Metric)	Baseline	+ DetRefiner	Gain
GLIP-Tiny (AP_r)	18.0	22.7	+4.7
Grounding DINO-Tiny (AP_r)	19.9	30.0	+10.1
MM-Grounding DINO-Base (AP_r)	35.4	41.6	+6.2
LLMDet-Large (AP_r)	43.9	50.1	+6.2

Generalization: consistent gains across other datasets

OV-COCO	Pascal VOC	ODinW13
GDINO-T (AP_{novel}): 57.4 → 58.1 +0.7	GDINO-T (AP): 56.3 → 60.0 +3.7	GLIP-T (mAP): 41.7 → 43.3 +1.6

Model Efficiency (on GDINO-T)

- Latency 324ms ⇒ 428ms
- FPS 3.08 ⇒ 2.33
- VRAM 4752MB ⇒ 6984MB

moderate overhead,
a single forward-pass refinement

Code and models

github.com/hitachi-rd-cv/detrefiner

44862 June 7, 2026 Soichiro Okazaki